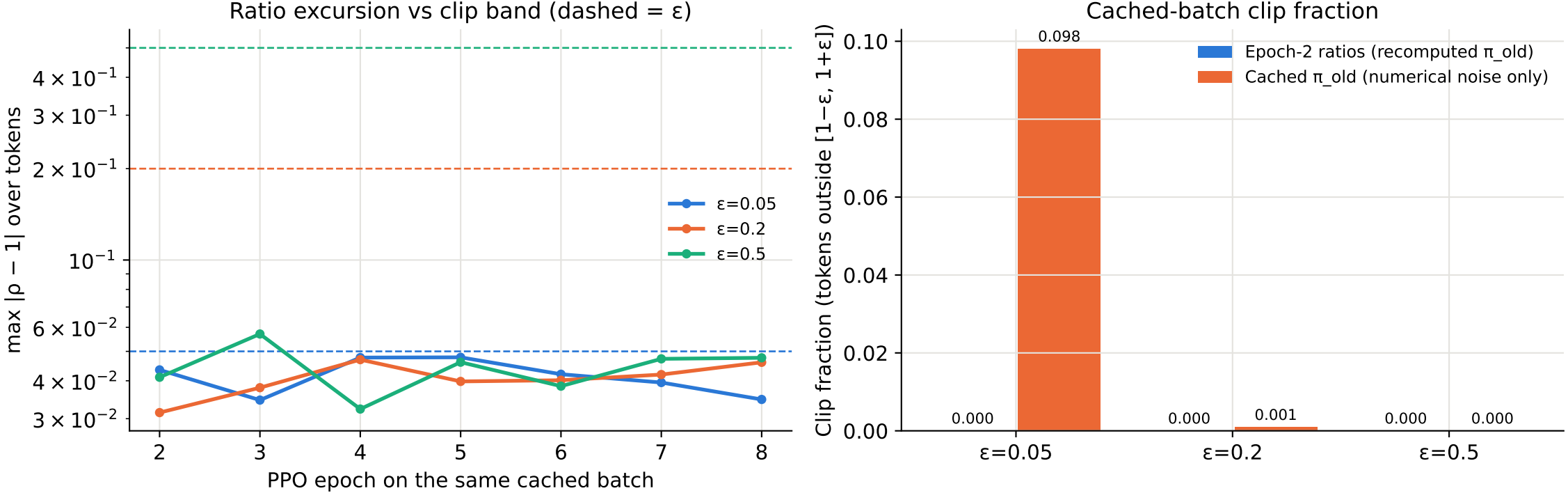
Ratio excursion vs clip band (dashed = ε)
max |ρ − 1| over tokens
PPO epoch on the same cached batch
ε=0.05
ε=0.2
ε=0.5
Cached-batch clip fraction
Clip fraction (tokens outside [1−ε, 1+ε])
Epoch-2 ratios (recomputed π_old)
Cached π_old (numerical noise only)
0.098
0.000
0.000
0.001
0.000
0.000
ε=0.05
ε=0.2
ε=0.5